Bank Customer Churn Prediction with Logistic Regression and Decision Tree

Abstract — The primary aim of the project detailed here was to conduct a comparative analysis of Logistic Regression and Decision Tree models for predicting customer churn from the Kaggle Bank Customer Churn dataset. The dataset is comprised of 10,000 customers and 10 features, plus one derived composite feature. Initial data exploration showed that the target variable was significantly imbalanced with 79.6% of customers staying and 20.4% leaving. The features Age, Geography, and Gender stood out as the ones with differences between the two categories. The two models were first assessed with the default settings and then tuned by cross-validation. Decision Tree with max_depth = 5 achieved better results than Logistic Regression after tuning (C = 0.20) in terms of all the metrics: accuracy (0.857 vs 0.808), F1 score (0.518 vs 0.194), precision (0.762 vs 0.561), and recall (0.392 vs 0.117). Besides, the Decision Tree overfitting issue revealed by default settings was fixed by tuning. These findings support the idea that a nonlinear model is more appropriate for the classification of customer churn. However, both models still struggle to identify the minority churned class.

Keywords: Customer Churn, Binary Classification, Logistic Regression, Decision Tree, Cross-Validation

I. Introduction

Customer churn, where customers stop using a company's services, is a major concern in the banking sector. Banks can save money by successfully identifying customers who are likely to leave, since retaining existing customers is generally less costly than acquiring new ones. Through predictive models built on past customer data, banks can effectively plan their retention approach and target their resources.

This study has focused on two supervised machine learning algorithms, Logistic Regression and Decision Tree, to predict if a given bank customer is likely to leave the bank using demographic and account data. Since the output variable Exited has only two categories (customer left or not), the problem is a binary classification task. Logistic Regression as a linear classifier predicts the probability of the positive class through a linear decision boundary. Decision Tree is a nonlinear classifier that sequentially divides the data space through simple decision rules based on the features. Thus, the linear and the nonlinear methods can be compared to understand which class of models better describes customer churn patterns.

The hypothesis of this paper is that customer demographic and account data contain information about churn and that a nonlinear model like Decision Tree will have higher performance than a linear one such as Logistic Regression due to the complex nature of customer behaviour. The data in question is the Kaggle Bank Customer Churn dataset [7] which has 10,000 customer records for 10 features such as age, geography, gender, balance, credit score, tenure, and activity status. The target variable Exited is an indication of whether a customer has left the bank (1) or not (0) and about 79.6% of the customers are in the stayed class while 20.4% in the churned class, so it is an imbalanced dataset. The objectives of this project are:

- Pre-process and explore the dataset using visualisation techniques to understand the distribution and relationships of features.
- Implement two supervised learning classification methods, Logistic Regression and Decision Tree, for the problem of customer churn prediction.
- Use appropriate performance metrics such as accuracy, confusion matrix, precision, recall and F1 score to assess and contrast the models.
- Understand which features are most influential in predicting customer churn.
- Determine whether a nonlinear model outperforms a linear model for this classification task.

II. Literature Review

Customer churn is a banking term for the event of a client switching to another bank or no

longer doing banking operations with a particular bank. From the bank's perspective, losing a customer represents a financial loss. Therefore, a lot of publications have focused on machine learning techniques for predicting customer churn. In this section, six studies are reviewed that are related to this project and cover various banking churn prediction scenarios.

Zhou and Huang [1] used a Kaggle bank churn dataset and tried multiple methods including Logistic Regression and Decision Tree. One of the main contributions of the work is the identification of the three most important features contributing to churn: customer age, number of products, and membership activity. This discovery offers a good starting point for the exploratory data analysis in the current project.

Huseyinov and Okocha [2] also used a Kaggle bank churn dataset. The main problem addressed in their study was imbalanced classes where the number of churned customers is outnumbered by the number of retained ones. The authors utilised SMOTE to create a balanced training set and discovered that none of the classifiers outperformed the others on the imbalanced data, whereas Random Forest was the best after the data was balanced. This is relevant to the current project, as it also deals with the imbalance where 20% are churned and 80% retained.

Kaur and Kaur [3] relied on Logistic Regression, Decision Tree, KNN and Random Forest, employing stratified sampling and 8-fold cross-validation for the banking churn problem. The use of the same two algorithms as the present work in the banking scenario confirms the appropriateness of these classifiers for such problems. Also, they used stratified splitting to guarantee that the class distribution remains constant in the train and test sets.

Nie et al. [4] did a head-to-head comparison of Logistic Regression and Decision Tree for credit card churn prediction based on a Chinese commercial bank dataset, so this is a very close study to the current project. The authors revealed that Logistic Regression had a lower average error rate than Decision Tree, meaning a linear classifier can sometimes perform better than a tree-based model depending on the nature of the data. This raises an important question for the current project: whether the same pattern will hold on the Kaggle dataset.

Gulati et al. [5] after adopting SMOTE for class balancing, found Decision Tree to be superior to Logistic Regression on their credit card churn data. This inconsistency in relative performance compared to Nie et al. [4] demonstrates that the results depend on the dataset and preprocessing steps. This further motivates the need for a direct comparison on the Churn_Modelling.csv dataset used in this project.

Lastly, Haarika and John [6] focused on the role of feature selection in banking churn prediction, by comparing different algorithms on a credit card churn dataset. It was demonstrated that it is possible to use only the most important features for training models and still get comparable accuracy to using all features, with the added advantage of shorter training time. This supports the preprocessing approach taken in the current project, where features such as RowNumber, CustomerId, and Surname have been removed as they do not carry any predictive power for customer churn.

III. Data Management

The dataset for the project is the Bank Customer Churn Modelling dataset, which was downloaded from Kaggle [7]. The dataset was downloaded as a CSV file named Churn_Modelling.csv which was then loaded into a Pandas DataFrame for the purpose of exploratory data analysis. The dataset contains 10,000 rows and 14 columns, where each row corresponds to a single bank customer and each column to a variable.

The first five rows of the dataset are displayed in Figure 1.

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

Figure 1: Sample of the first five rows of the dataset

The original set of 14 columns is comprised of a mixture of numerical, categorical, and binary variables. CreditScore, Age, Tenure, Balance, NumOfProducts, and EstimatedSalary are numerical variables that characterise the customer's demographics and account activity. Geography and Gender are categorical variables stored as object data types, representing the customer's country and gender respectively. HasCrCard and IsActiveMember are binary variables encoded as integers, where 1 indicates yes and 0 indicates no. The target variable is Exited, which indicates whether the customer left the bank (1) or stayed (0). The remaining three columns are RowNumber, CustomerId, and Surname.

The quality of the data was checked before proceeding with any analysis. There were no missing values found in any of the 14 columns, and no duplicated rows were present in the dataset. This means the data did not require any imputation or row removal for quality purposes.

However, not all columns are useful for predicting churn. RowNumber is just a row index that does not carry any information about customer behaviour. CustomerId has 10,000 unique values, one for each row, confirming that it serves only as a label. Surname is a personal name that has nothing to do with whether a customer will leave the bank. Since the inclusion of these columns would only add noise to the models, all three were dropped from the dataset, reducing the number of columns from 14 to 11, leaving 10 feature variables and the target variable Exited, before any further analysis.

IV. Exploratory Data Analysis

Before building any models, the dataset was explored to gain a better understanding of how variables are distributed and to identify patterns that could help predict churn. The target variable Exited shows a class imbalance pattern since there are 7,963 customers (79.6%) in class 0 (stayed) and only 2,037 customers (20.4%) in class 1 (churned), which is illustrated by Figure 2.

The significance of this imbalance lies in the fact that a model simply predicting all customers as stayed would have a 79.6% accuracy without learning any meaningful patterns. Therefore, metrics other than accuracy such as precision, recall, and F1 score are also required when evaluating model performance.

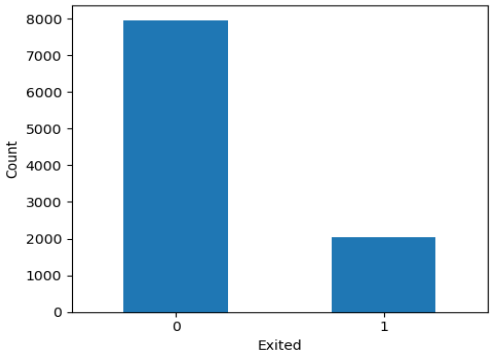


Figure 2: Distribution of the target variable Exited

In order to check if there is any association between categorical features and churn, normalised crosstabs were used to compare the churn rates across categories. As it can be seen, the churn rate for Germany is significantly higher (32.4%) than for France (16.2%) and Spain (16.7%), which suggests that Geography may contain information useful for prediction. Similarly, female customers are more likely to churn (25.1%) than male customers (16.5%), thus Gender is another feature that could be predictive. These comparisons are plotted in Figure 3.

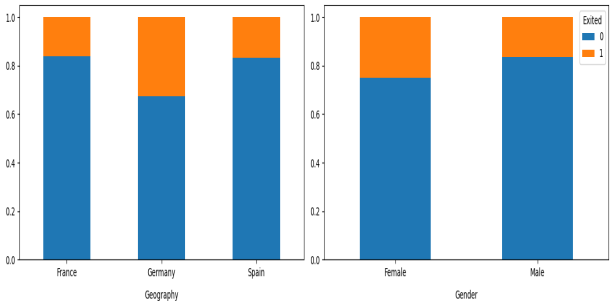


Figure 3: Normalised churn rates by Geography and Gender

The shapes and spreads of Age and Balance were analysed with histograms, density plots, and boxplots. As illustrated in Figure 4, the Age distribution is moderately skewed to the right and there are a few outliers at the higher values. Since these older customers are real cases in a banking dataset, all the outliers were kept in the model.

On the other hand, Balance has many customers with zero balance leading to a distribution consisting of two separate groups.

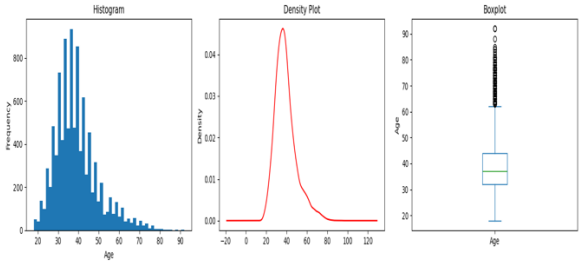


Figure 4: Distribution of Age (histogram, density plot, and boxplot)

Comparing Age between churned and stayed customers, Figure 5 demonstrates a notable difference: churned customers are generally older as their median age is greater than that of stayed customers. This remains in accordance with the work of Zhou and Huang [1] who identified Age as one of the major components determining churn. By splitting the Exited variable groupings, the Balance distributions are also distinguishable, since the average balance of churned customers is higher than that of stayed customers.

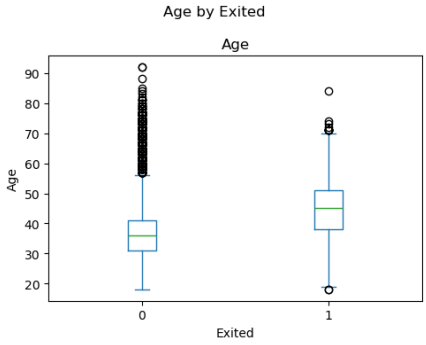


Figure 5: Age distribution grouped by Exited

To sum up, the exploratory data analysis indicates that Age, Geography, and Gender are the most promising features in terms of churn prediction, while the target class imbalance should be accounted for when model evaluation is performed.

V. Methodology

Before applying any machine learning models, the remaining preprocessing steps were completed. Geography and Gender are text features that cannot be used directly by machine learning algorithms, so both were converted to integer codes using category codes. A new feature, BalanceSalaryRatio, was derived by dividing Balance by EstimatedSalary, with 1 added to the denominator to avoid division by zero in order to capture the interaction between two variables because the balance relative to the salary is probably more informative than each variable taken separately.

To identify features that possibly contain the same information, a correlation matrix was calculated and displayed as a heatmap in Figure 6.

There are no features that show high correlation with each other, confirming that all features provide distinct information and none need to be dropped. The correlation values with the target variable Exited show that Age has the strongest positive correlation (0.285), followed by Balance (0.119), while IsActiveMember has the strongest negative correlation (−0.156).



Figure 6: Correlation heatmap

To determine which features have the highest potential to predict the outcome, four different feature selection techniques were used: chi-squared test, F-test, mutual information, and ExtraTreesClassifier. As can be seen from Figure 7, the weakest feature varies depending on the method used— chi-squared and F-test identify HasCrCard as the weakest, while mutual information and ExtraTreesClassifier rank other features lowest. Since no single feature is consistently ranked as weak across all four methods, all 11 features (10 original plus the composite BalanceSalaryRatio) were retained for modelling.

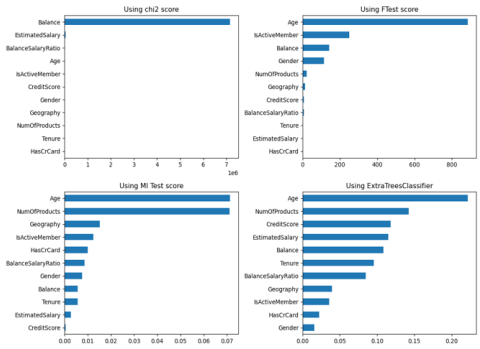


Figure 7: Feature selection scores from four methods

The features were further processed by scaling them to a 0–1 range with the use of MinMaxScaler. This is particularly important for Logistic Regression, as variables with larger ranges such as EstimatedSalary could otherwise dominate the learning of weights. Decision Trees are not affected by feature scaling since they split based on thresholds rather than distances, but scaling was applied to both models for consistency.

The project considers two supervised classification algorithms: Logistic Regression and Decision Tree. Logistic Regression is a linear method and it expresses the probability of churn as a linear function of the features, thus it is well suited for problems where the relationship between features and the target is approximately linear. Decision Tree is a nonlinear method that splits the data through a series of decisions based on feature values, enabling it to capture more complex patterns and interactions between features. Comparing a linear and a nonlinear approach provides insight into the nature of the churn prediction problem. Both algorithms are commonly used in the churn prediction literature [1]–[6].

To evaluate the models, the data was divided into training and test sets, 80% training and 20% test. Both models were trained and evaluated initially with their default parameters using 5-fold cross-validation to get default performance metrics. After this, hyperparameters were tuned using 10-fold cross-validation on the training data: the regularisation strength C was varied for Logistic Regression, and the maximum depth (max_depth) was varied for Decision Tree. The best hyperparameter for each model was selected based on the highest cross-validation accuracy, and final performance was evaluated on the test set using accuracy, precision, recall, F1 score, and confusion matrix.

VI. Results and Analysis

The performances of the models with default parameters were compared first and it was found that Logistic Regression with default parameters gave a test accuracy of 0.816 whereas the recall was very low, only 0.170, meaning that only 17% of the customers who left were correctly identified. On the other hand, Decision Tree achieved a lower test accuracy of 0.789 but a much higher recall of 0.499. However, Decision Tree showed severe overfitting, with a training accuracy of 1.000 compared to its test accuracy, indicating that the model had memorised the training data rather than learning general patterns. The top levels of the default Decision Tree are visualised in Figure 8.

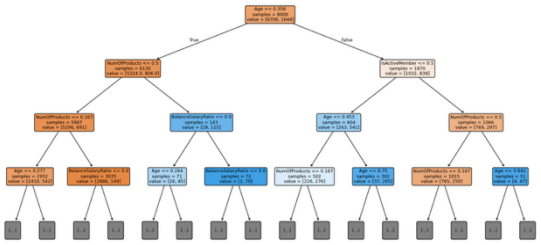


Figure 8: Top levels of the default Decision Tree

These default results motivated the need for hyperparameter tuning. Following tuning with 10-fold cross-validation, the best regularisation strength for Logistic Regression was C = 0.20, as shown in Figure 9, and the best maximum depth for Decision Tree was max_depth = 5, as shown in Figure 10.

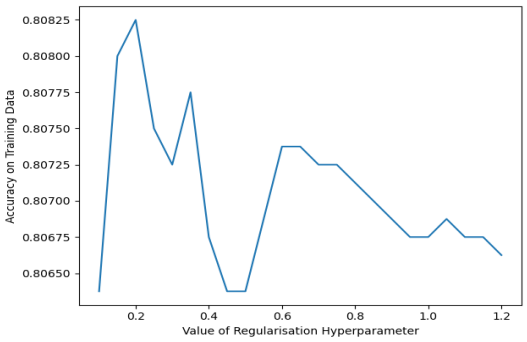


Figure 9: Cross-validation accuracy across regularisation strength C for Logistic Regression

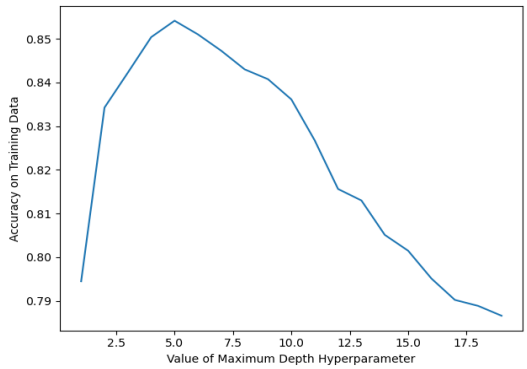


Figure 10: Cross-validation accuracy across maximum depth for Decision Tree

The tuned models were then evaluated on the test set, with confusion matrices shown in Figure 11 and Figure 12.

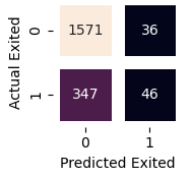


Figure 11: Confusion matrix for tuned Logistic Regression

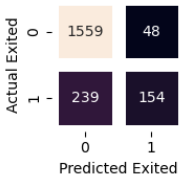


Figure 12: Confusion matrix for tuned Decision Tree

The tuned Logistic Regression achieved a test accuracy of 0.808, F1 score of 0.194, precision of 0.561, and recall of 0.117. Its confusion matrix shows 1,571 correct stayed predictions and 46 correct churned predictions, but 347 churned customers were misclassified as stayed. This means the model misses the majority of churned customers, which limits its usefulness for a bank trying to identify customers at risk of leaving.

The tuned Decision Tree achieved a test accuracy of 0.857, F1 score of 0.518, precision of 0.762, and recall of 0.392. Its confusion matrix shows 1,559 correct stayed predictions and 154 correct churned predictions, with 239 churned customers misclassified as stayed. While still missing some churned customers, Decision Tree identifies significantly more than Logistic Regression.

The difference in training and test accuracy is near-zero for Logistic Regression because the training accuracy is 0.809 and the test accuracy is 0.808, thus no overfitting is indicated, whereas Decision Tree differs by only 0.002 between training accuracy of 0.859 and test accuracy of 0.857. Compared to the default Decision Tree where training accuracy was 1.000, tuning max_depth to 5 effectively resolved the overfitting problem.

Decision Tree with max_depth = 5 outperforms Logistic Regression (C = 0.20) across all test metrics including accuracy, F1 score, precision, and recall. This finding is consistent with the hypothesis made in the introduction that a nonlinear model would outperform a linear model for this churn prediction task. Besides the regularisation feature of Logistic Regression that limits overfitting as demonstrated by the near-zero difference between its training and test accuracy, the interpretability of the model is another advantage that can be taken into account when assessing the model. However, its linear decision boundary limits its ability to capture complex interactions between features, which contributes to its poor recall on the minority class.

On the contrary, Decision Tree explores nonlinear relationships and interactions between features through its splitting structure, which explains its stronger performance across all metrics. The major drawback of Decision Tree is the tendency to overfit when it has no depth limitation, however this was effectively controlled by limiting the tree depth to 5.

VII. Conclusion

This project made use of Logistic Regression and Decision Tree to predict whether a customer will leave the bank or not based on the

Kaggle Bank Customer Churn dataset containing 10,000 customer records and 10 features, plus the derived BalanceSalaryRatio. After hyperparameter tuning through 10-fold cross-validation, Decision Tree with max_depth = 5 outperformed Logistic Regression (C = 0.20) across all metrics: accuracy (0.857 vs 0.808), F1 score (0.518 vs 0.194), precision (0.762 vs 0.561), and recall (0.392 vs 0.117). Tuning also resolved the overfitting observed in the default Decision Tree, reducing the training-test accuracy difference from 0.211 to 0.002.

These findings support the hypothesis that a nonlinear model outperforms a linear model for this task, as it captures more complex patterns in the data. Major accomplishments include identifying Age, Geography, and Gender as the most informative features, showing that hyperparameter tuning resolves overfitting, and emphasising the need for using several metrics when dealing with imbalanced datasets.

The target variable has class imbalance where only 20.4% of the total customers have churned, hence the models have become biased towards the majority class resulting in low recall. Also, determining the right hyperparameter values was challenging because increasing Decision Tree depth beyond 5 led to overfitting.

In terms of business use, the tuned Decision Tree can be used as an early warning system to identify the customers most likely to leave, enabling banks to target them with retention offers, which is more cost-effective than acquiring new customers.

Nevertheless, both models still find it hard to deal with the minority class. As future work, SMOTE might be implemented as a solution for class imbalance, as demonstrated by Huseyinov and Okocha [2] and Gulati et al. [5]. Other possibilities include ensemble methods such as Random Forest or Gradient Boosting, and additional composite variables such as combinations of Age and Geography.

VIII. References

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